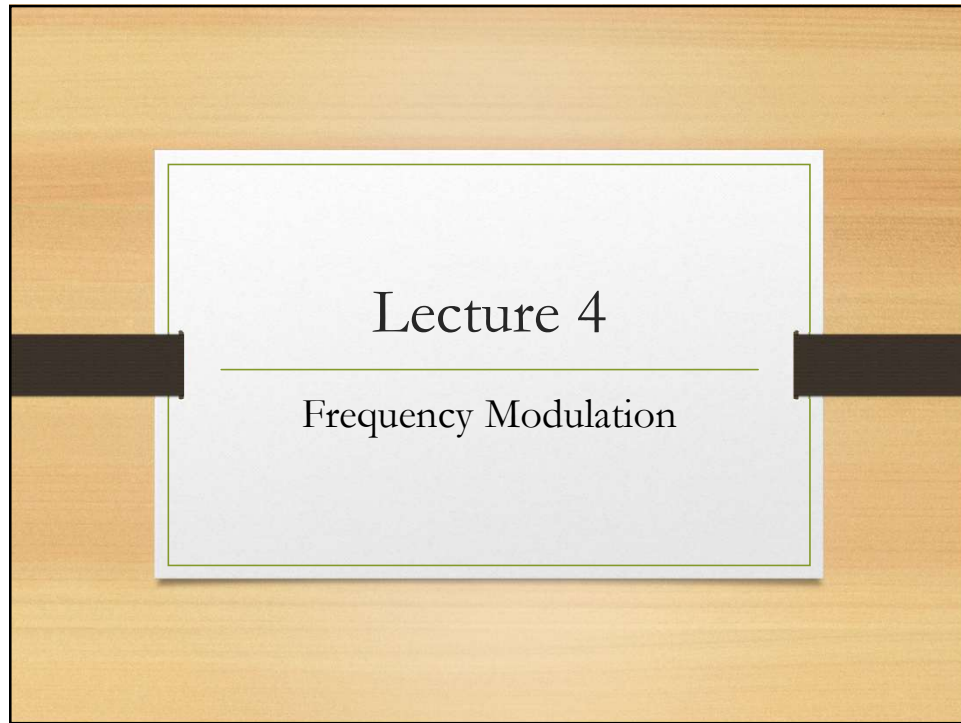
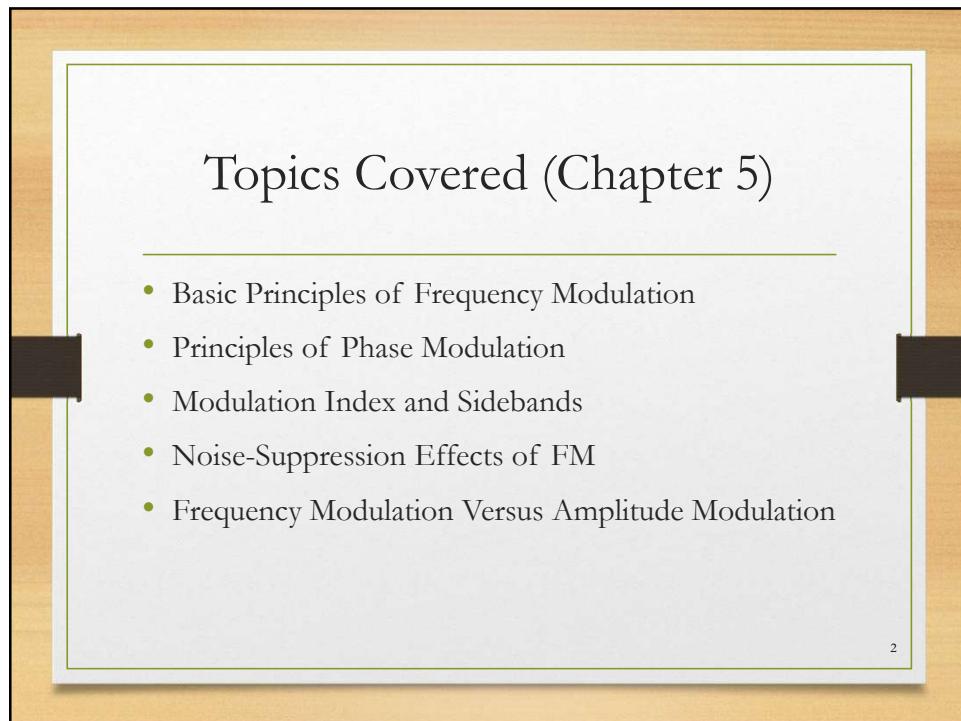




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5-1: Basic Principles of Frequency Modulation

- A sine wave carrier can be modified for the purpose of transmitting information from one place to another by varying its frequency. This is known as **frequency modulation (FM)**.
- In FM, the carrier amplitude remains constant and the carrier frequency is changed by the modulating signal.

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5-1: Basic Principles of Frequency Modulation

- As the amplitude of the information signal varies, the carrier frequency shifts proportionately.
- As the modulating signal amplitude increases, the carrier frequency increases.
- With no modulation the carrier is at its normal center or **resting frequency**.

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Instantaneous Frequency

- The instantaneous frequency (f_i) of the carrier is given by $f_i = f_c + k_f v_m(t)$
- Frequency sensitivity (k_f) and is the change in carrier frequency per volt of modulating signal.
- Frequency swing (f_{p-p}) is the difference between the minimum instantaneous frequency $f_{i_{min}}$ and the maximum instantaneous frequency $f_{i_{max}}$.
 - $f_{p-p} = f_{i_{max}} - f_{i_{min}} = k_f(v_{max} - v_{min}) = k_f v_{m-p-p}$

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Frequency Deviation

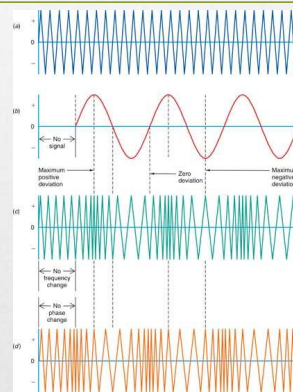
- Frequency deviation (f_d) is the maximum amount by which the carrier deviates from the unmodulated carrier value.
 - Frequency deviation = $\frac{1}{2}$ (Frequency swing)
- The maximum allowable (by regulation) frequency deviation is known as the rated system deviation.
 - Used to limit the bandwidth utilisation.
 - Typical values –
 - 75 kHz for FM radio
 - 25 kHz for TV sound
 - 5 kHz for 2-way radio.

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5-1: Basic Principles of Frequency Modulation

Figure 5-1: FM and PM signals. The carrier is drawn as a triangular wave for simplicity, but in practice it is a sine wave. (a) Carrier. (b) Modulating signal. (c) FM signal. (d) PM signal.



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5-2: Principles of Phase Modulation

- When the amount of phase shift of a constant-frequency carrier is varied in accordance with a modulating signal, the resulting output is a **phase-modulation (PM) signal**.
- Phase modulators produce a phase shift which is a time separation between two sine waves of the same frequency.
- The greater the amplitude of the modulating signal, the greater the phase shift.

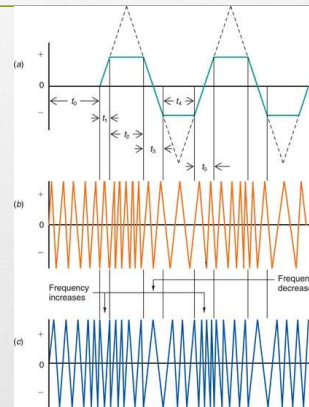
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5-2: Principles of Phase Modulation

- The maximum frequency deviation produced by a phase modulator occurs during the time that the modulating signal is changing at its most rapid rate.

Figure 5-3: A frequency shift occurs in PM only when the modulating signal amplitude varies. (a) Modulating signal. (b) FM signal. (c) PM signal.



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Modulating Signal and Carrier Deviation

- In FM and in PM, the frequency deviation is directly proportional to the amplitude of the modulating signal.
- In PM, the maximum amount of leading or lagging phase shift occurs at the peak amplitudes of the modulating signal.
- In PM, the carrier deviation is proportional to both the modulating frequency and the amplitude.

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5-2: Principles of Phase Modulation

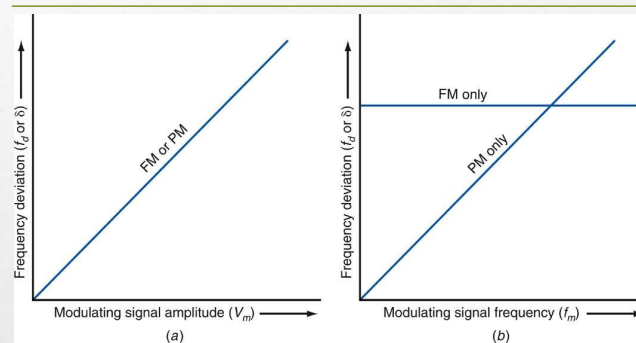


Figure 5-4: Frequency deviation as a function of (a) modulating signal amplitude and (b) modulating signal frequency.

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5-2: Principles of Phase Modulation

Converting PM into FM

- In order to make PM compatible with FM, the deviation produced by frequency variations in the modulating signal must be compensated for.
- This compensation can be accomplished by passing the intelligence signal through a low-pass RC network.
- This RC low-pass filter is called a **frequency-correcting network, predistorter, or 1/f filter** and causes the higher modulating frequencies to be attenuated.
- The FM produced by a phase modulator is called **indirect FM**.

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5-3: Modulation Index and Sidebands

- Any modulation process produces sidebands.
- When a constant-frequency sine wave modulates a carrier, two side frequencies are produced.
- Side frequencies are the sum and difference of the carrier and modulating frequency.
- The bandwidth of an FM signal is usually much wider than that of an AM signal with the same modulating signal.

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Modulation Index

- β (also known as m_f) is the ratio of frequency deviation of the carrier to the frequency of the modulating signal.

$$\beta = \frac{f_d}{f_m}$$

- β is a non-integer number that can have a large value (eg. 25).
- β has a single value only when a test-tone is used.
- For speech and music, β changes constantly. In this case it is more convenient to use Deviation ratio D (maximum values)

$$D = \frac{f_d}{f_{max}}$$

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NBFM ($\beta \leq 0.25$)

- Mathematically, for a modulating signal $v_m(t) = V_m \cos(2\pi f_m t)$

$$v_{fm}(t) = V_c \cos[2\pi f_c t + \beta \sin(2\pi f_m t)]$$

- The solution of this equation depends on the value of β

$$v_{NBFM}(t) = V_c \cos(2\pi f_c t) + \frac{\beta V_c}{2} \cos[2\pi(f_c - f_m)t + \pi] + \frac{\beta V_c}{2} \cos[2\pi(f_c + f_m)t]$$

- If β small (≤ 0.25) then approximations $[\sin\theta \approx \theta, \cos\theta \approx 1]$ lead to NBFM
- A carrier and two side frequencies – similar to AM

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WBFM ($\beta > 0.25$)

- When β exceeds 0.25 then simplifying approximations cannot be used.
- Use Bessel functions.
- Result show a carrier and an infinite number of side frequencies spaced at multiples of the modulating frequency, f_m , above and below the carrier.
- The amplitude of these frequencies depend on the value of β

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5-3: Modulation Index and Sidebands

Bessel Functions

- The equation that expresses the phase angle in terms of the sine wave modulating signal is solved with a complex mathematical process known as **Bessel functions**.
- Bessel coefficients are widely available and it is not necessary to memorize or calculate them.

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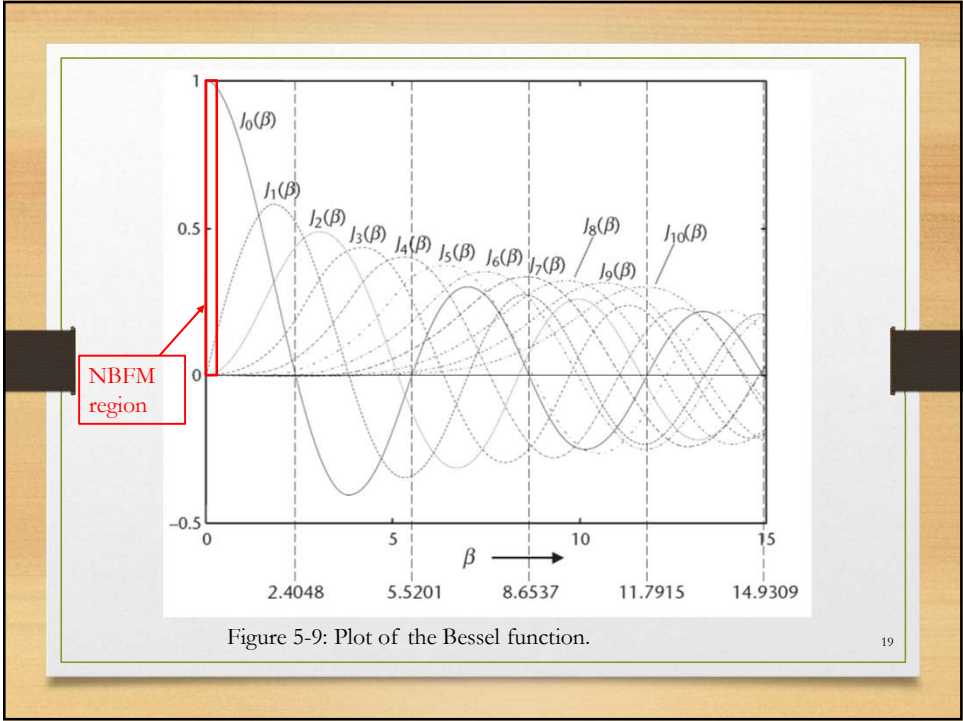
5-3: Modulation Index and Sidebands

Modulation Index	Carrier	Sidebands (Pairs)															
		1st	2d	3d	4th	5th	6th	7th	8th	9th	10th	11th	12th	13th	14th	15th	16th
0.00	1.00	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
0.25	0.98	0.12	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
0.5	0.94	0.24	0.03	—	—	—	—	—	—	—	—	—	—	—	—	—	—
1.0	0.77	0.44	0.11	0.02	—	—	—	—	—	—	—	—	—	—	—	—	—
1.5	0.51	0.56	0.23	0.06	0.01	—	—	—	—	—	—	—	—	—	—	—	—
2.0	0.22	0.58	0.35	0.13	0.03	—	—	—	—	—	—	—	—	—	—	—	—
2.5	-0.05	0.50	0.45	0.22	0.07	0.02	—	—	—	—	—	—	—	—	—	—	—
3.0	-0.26	0.34	0.49	0.31	0.13	0.04	0.01	—	—	—	—	—	—	—	—	—	—
4.0	-0.40	-0.07	0.36	0.43	0.28	0.13	0.05	0.02	—	—	—	—	—	—	—	—	—
5.0	-0.18	-0.33	0.05	0.36	0.39	0.26	0.13	0.05	0.02	—	—	—	—	—	—	—	—
6.0	0.15	-0.28	-0.24	0.11	0.36	0.36	0.25	0.13	0.06	0.02	—	—	—	—	—	—	—
7.0	0.30	0.00	-0.30	-0.17	0.16	0.35	0.34	0.23	0.13	0.06	0.02	—	—	—	—	—	—
8.0	0.17	0.23	-0.11	-0.29	-0.10	0.19	0.34	0.32	0.22	0.13	0.06	0.03	—	—	—	—	—
9.0	-0.09	0.24	0.14	-0.18	-0.27	-0.06	0.20	0.33	0.30	0.21	0.12	0.06	0.03	0.01	—	—	—
10.0	-0.25	0.04	0.25	0.06	-0.22	-0.23	-0.01	0.22	0.31	0.29	0.20	0.12	0.06	0.03	0.01	—	—
12.0	-0.05	-0.22	-0.08	0.20	0.18	-0.07	-0.24	-0.17	0.05	0.23	0.30	0.27	0.20	0.12	0.07	0.03	0.01
15.0	-0.01	0.21	0.04	0.19	-0.12	0.13	0.21	0.03	-0.17	-0.22	-0.09	0.10	0.24	0.28	0.25	0.18	0.12

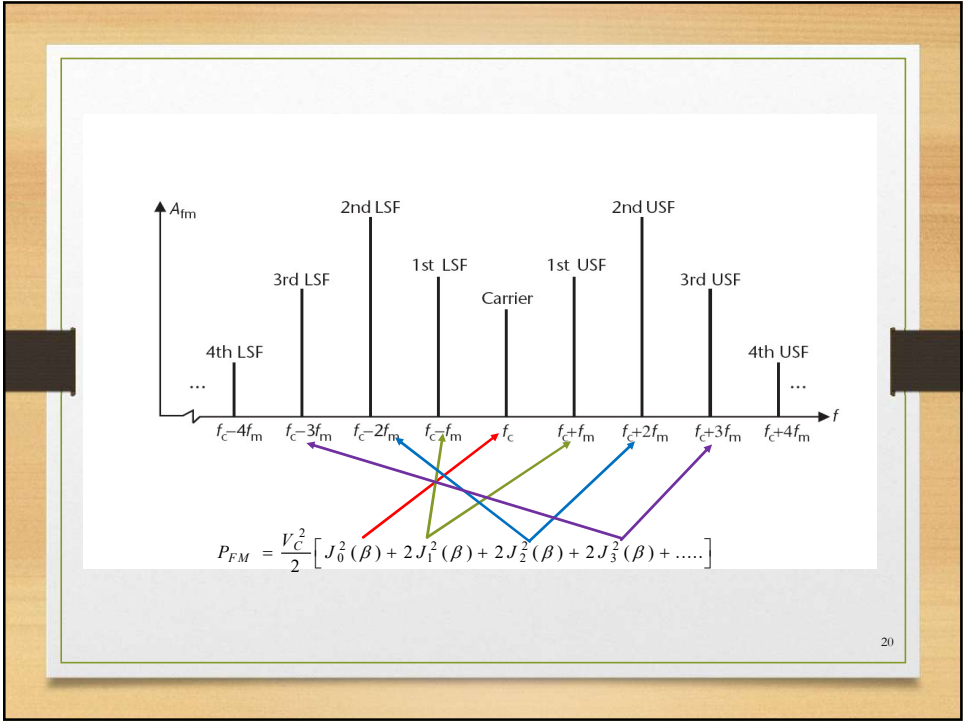
Figure 5-8: Carrier and sideband amplitudes for different modulation indexes of FM signals based on the Bessel functions.

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FM Signal Power

- The power in the FM signal depends only on the amplitude and is independent of the frequency and phase. The power is equal to the power in the unmodulated carrier

$$P_{FM} = \frac{1}{2} V_C^2$$

- The total power in the FM signal can be obtained by summing the power in all the spectral components of the FM signal.

$$P_{FM} = \frac{V_C^2}{2} [J_0^2(\beta) + 2J_1^2(\beta) + 2J_2^2(\beta) + 2J_3^2(\beta) + \dots]$$

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FM Signal Bandwidth

- The higher the modulation index in FM, the greater the number of significant sidebands and the wider the bandwidth of the signal.
- When spectrum conservation is necessary, the bandwidth of an FM signal can be restricted by putting an upper limit on the modulation index.

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FM Signal Bandwidth

- Example:**

If the highest modulating frequency is 3 kHz and the maximum deviation is 6 kHz, what is the modulation index?

$$m_f = 6 \text{ kHz} / 3 \text{ kHz} = 2$$

What is the bandwidth?

$$BW = 2f_m N$$

Where N is the number of significant* sidebands

$$BW = 2(3 \text{ kHz})(4) = 24 \text{ kHz}$$

*Significant sidebands are those that have an amplitude of greater than 1% (.01) in the Bessel table.

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Carson's Bandwidth

- Carson's bandwidth is an empirical definition which gives the width of the band of frequencies centred at f_c that contains at least 98% of the power.

$$\begin{aligned} \text{Bandwidth} &= 2(\beta + 1)f_m \\ &= 2(f_D + f_m) \end{aligned}$$

- This formula effectively encompasses all the side-frequencies that exceed 10% of the relative amplitude of the unmodulated carrier.

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1% Bandwidth

- Is the bandwidth from the (n_{MAX}) th. LSF to the (n_{MAX}) th. USF beyond which the amplitude of every spectral component is less than 1% of the unmodulated carrier amplitude.
- Results are usually read from the Bessel table.
 - 1% bandwidth accounts for 99.97% of the total FM power.
 - Carson bandwidth accounts of 98.4% of the total FM power.
 - Generally use Carson bandwidth because it is easier.

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FM Versus AM

Advantages of FM

- FM typically offers some significant benefits over AM.
 - FM has superior immunity to noise, made possible by clipper limiter circuits in the receiver.
 - In FM, interfering signals on the same frequency are rejected. This is known as the **capture effect**.
 - FM signals have a constant amplitude and there is no need to use linear amplifiers to increase power levels. This increases transmitter efficiency.

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5-5: Frequency Modulation Versus Amplitude Modulation

Disadvantages of FM

- FM uses considerably more frequency spectrum space.
- FM has used more complex circuitry for modulation and demodulation.
- In the past, the circuits used for frequency modulation and demodulation involved were complex. With the proliferation of ICs, complex circuitry used in FM has all but disappeared. ICs are inexpensive and easy to use. FM and PM have become the most widely used modulation method in electronic communication today.

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5-5: Frequency Modulation Versus Amplitude Modulation

Major applications of AM and FM

Application	Type of Modulation
AM broadcast radio	AM
FM broadcast radio	FM
FM stereo multiplex sound	DSB (AM) and FM
TV sound	FM
TV picture (video)	AM, VSB
TV color signals	Quadrature DSB (AM)
Cellular telephone	FM, FSK, PSK
Cordless telephone	FM, PSK
Fax machine	FM, QAM (AM plus PSK)
Aircraft radio	AM
Marine radio	FM and SSB (AM)
Mobile and handheld radio	FM
Citizens band radio	AM and SSB (AM)
Amateur radio	FM and SSB (AM)
Computer modems	FSK, PSK, QAM (AM plus PSK)
Garage door opener	OOK
TV remote control	OOK
VCR	FM
Family Radio service	FM

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